

ME597: Machine Learning (ML) Tutorial 3

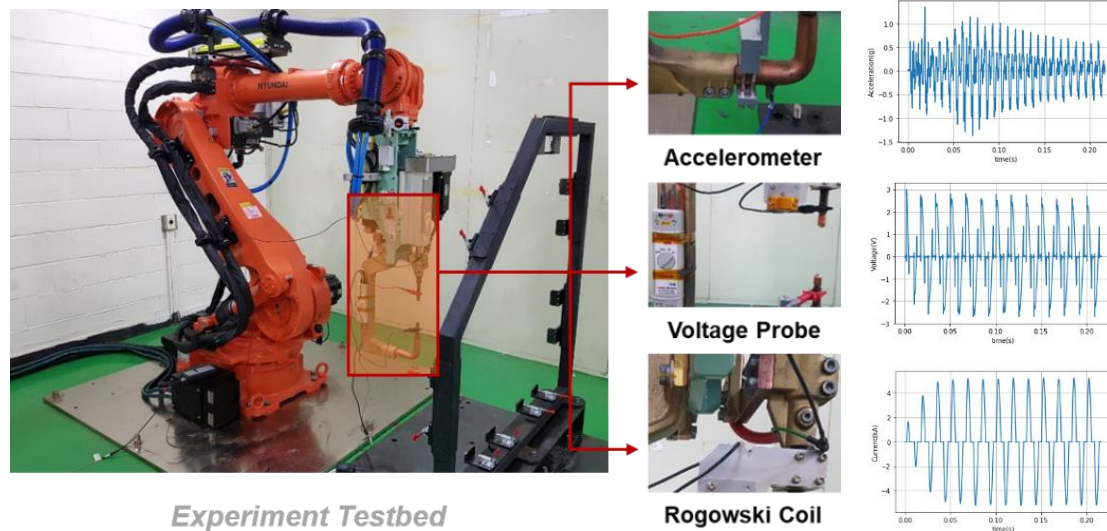
Week 2

School of Mechanical Engineering
Purdue University

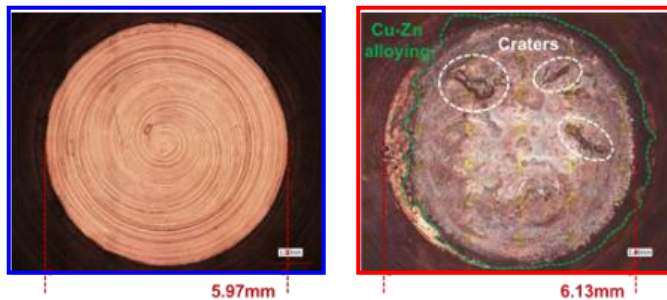
Dr. Martin Jun

Dataset for ML Tutorial

- Robotic spot-welding process data



Experiment Testbed



Welding electrode tip (Normal / Abnormal)

Sampling Data	Acceleration
	Current
	Voltage
Sampling freq.	12,800 Hz
Sampling time	0.2167 seconds (2,774 samples for 1 Data)

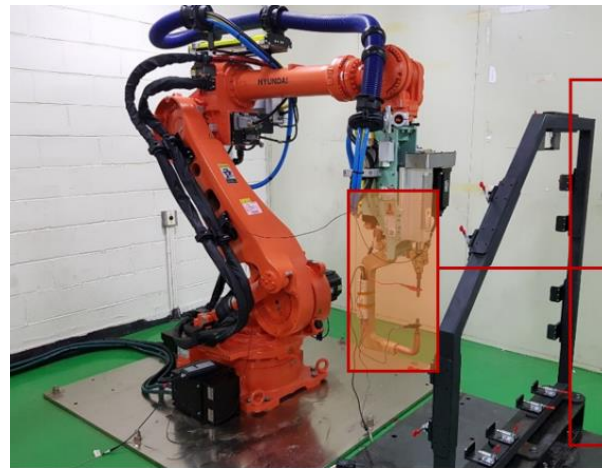
Sensor data types & sampling condition

Robotic spot-welding process demonstration



Dataset for ML Tutorial

- Robotic spot-welding process data



Experiment Testbed



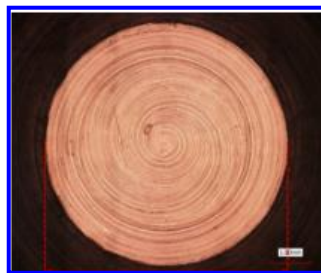
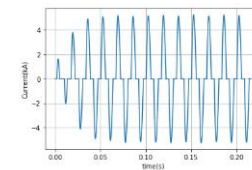
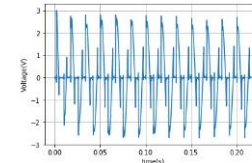
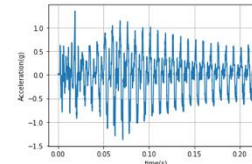
Accelerometer



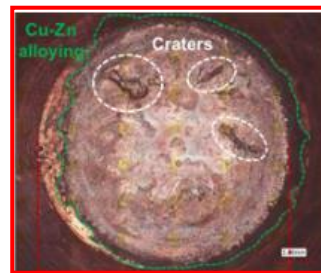
Voltage Probe



Rogowski Coil



5.97mm



6.13mm

Welding electrode tip (Normal / Abnormal)

Sampling Data	Acceleration
	Current
	Voltage
Sampling freq.	12,800 Hz
Sampling time	0.2167 seconds (2,774 samples for 1 Data)

Sensor data types & sampling condition

	time	ACC	Voltage	Current
0	0.000000	0.016440	-0.014200	0.001433
1	0.000078	0.013822	-0.015707	0.001504
2	0.000156	0.019431	-0.016963	0.001433
3	0.000234	0.003726	-0.017248	0.001549
4	0.000313	0.022049	-0.009840	0.001864
...	...	...	...	...
2769	0.216330	0.077392	0.002404	0.003453
2770	0.216410	0.030276	0.003952	0.003498
2771	0.216480	0.017188	0.005504	0.003397
2772	0.216560	0.037754	-0.000108	0.003506
2773	0.216640	-0.016093	-0.001652	0.003600

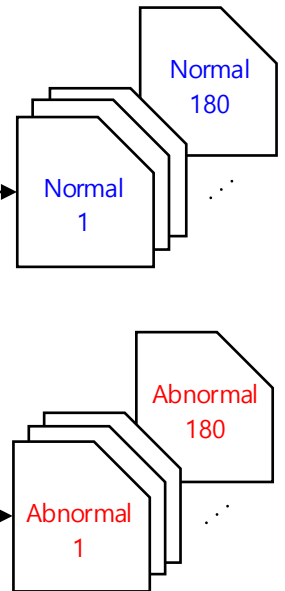
2774 rows × 4 columns

Example Data

(csv file)

360 Data

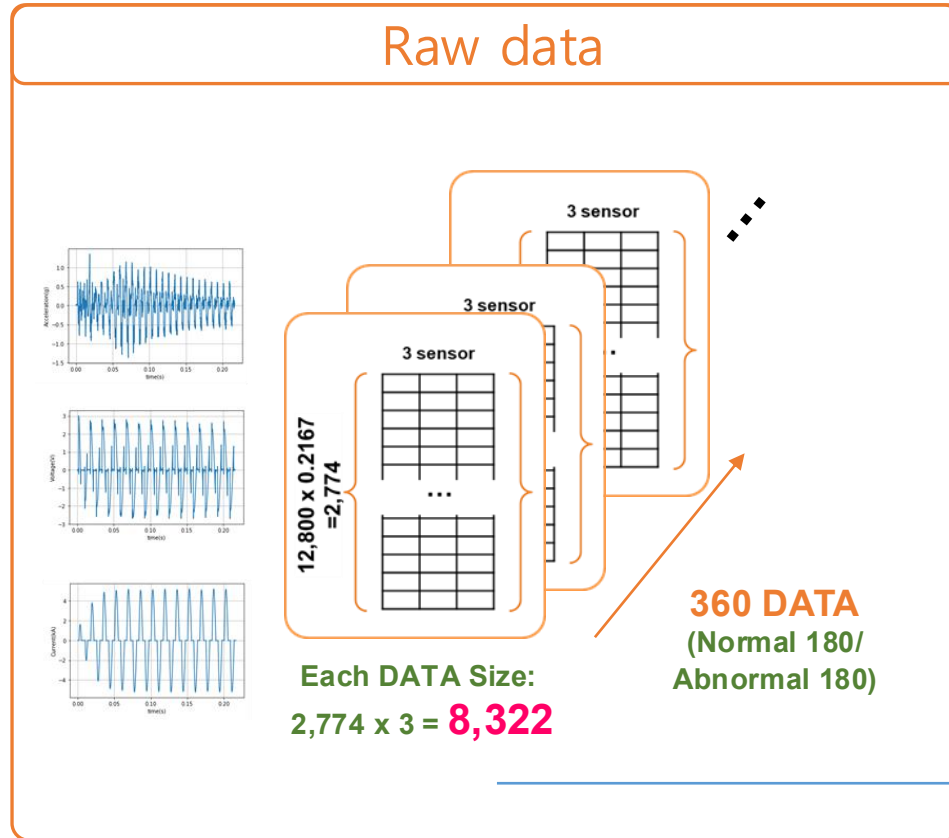
(Normal 180 /
Abnormal 180)



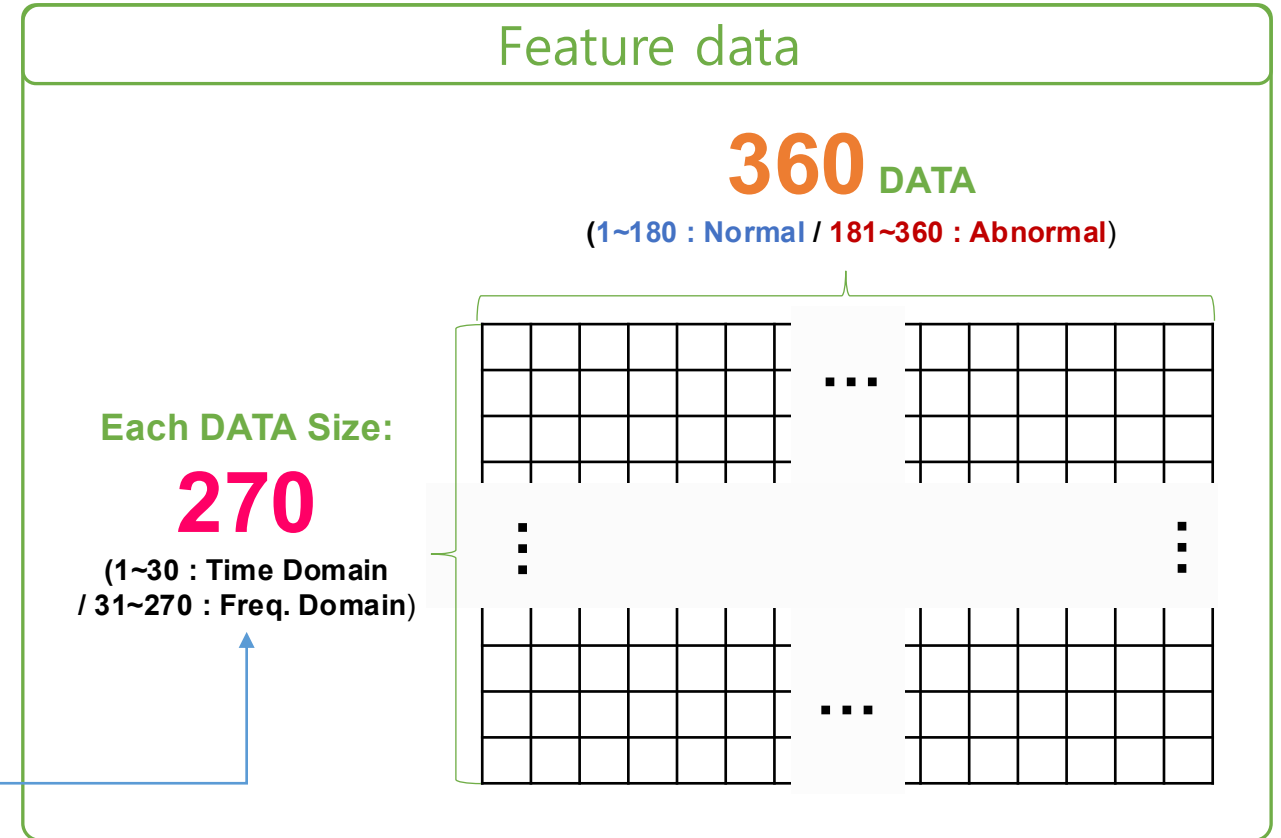
PURDUE **MECHANICAL
ENGINEERING**



Feature Extraction



360 Data files
(.csv)



1 Data files
(.csv)